Prime Constellations of $Q(n) = n^q - (n-1)^q$: A Bateman–Horn Census to 8×10^{11}, Certified 1000-Digit Quadruplets, and a 200,000-Digit Q_{47} Probable Prime

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Abstract

We study prime values and prime constellations of $Q_q(n) = n^q - (n-1)^q$, the degree-$(q-1)$ analogues of the cuban primes, with emphasis on $q = 47$. Every prime factor of $Q_q(n)$ is $\equiv 1 \pmod{2q}$, and each such prime excludes exactly $q-1$ residue classes of n; from this we compute the Bateman–Horn constants $C_k(q)$ of the k-tuples $Q_q(n), \ldots, Q_q(n+k-1)$ for ten exponents. The data agree with Bateman–Horn at every scale we tested. For Q_{47} we count 18,473,624 primes for $n \le 2 \times 10^9$ (predicted 18,464,550), 2,393 prime quadruplets for $n < 8 \times 10^{11}$ (predicted 2,404.3) and 21 quintuplets (predicted 17.9). In windows at $n \approx 10^{20}$ and $n \approx 4.8 \times 10^{21}$ we find quadruplets of 922 and 1000 digits; for eight of the 1000-digit quadruplets all four members carry ECPP primality certificates. A sieve-and-PRP search at 200,000 digits found

$$Q_{47}(586135293385 \cdot 10^{4336} + 1647),$$

a probable prime of exactly 200,000 digits that passes Fermat tests to bases 3, 5, 7, 11 and independent strong base-11 and BPSW tests with GMP; we believe it is the largest known prime-candidate value of Q_{47}. The large differences between exponents in quadruplet frequency are explained by the constants $C_4(q)$, which are governed by the smallest prime $p \equiv 1 \pmod{q}$: when $2q+1$ is prime, that single prime cuts C_4 by an order of magnitude. Whether q is a safe prime plays no role. A previously reported correlation between quadruplet positions and Riemann zeros disappears completely after detrending.

Keywords: prime-generating polynomials, prime constellations, Bateman–Horn conjecture, singular series, cuban primes, probable primes, ECPP, PFGW, Primo
MSC 2020: 11N32, 11A41, 11Y11, 11N35, 11N05

Changes from Version 1

Version 1 (January 2026) reported 15 quadruplets of Q_{47} below 2×10^9, attributed their abundance to $q = 47$ being a safe prime through an "interference potential" $I(q) = d(q-1) - 2$, and reported a correlation $r = 0.994$ between quadruplet positions and Riemann zeros. Version 2 replaces that framework with an explicit Bateman–Horn analysis:

- the constants $C_k(q)$, $k \le 6$, are computed (Section 3); they predict the observed counts, and the variation between exponents comes from the smallest prime $p \equiv 1 \pmod{q}$, not from the safe-prime property (Section 7);
- the Q_{47} census is extended to 8×10^{11}, with quintuplets (Section 6);

- quadruplets at 922 and 1000 digits, with ECPP certificates, and a 200,000-digit probable prime are added (Sections 8, 9);
- the Riemann-zero correlation is re-examined with a null model and a detrended test, and is not significant (Section 10);
- the subgroup lattice of $(\mathbb{Z}/47\mathbb{Z})^*$, drawn in v1 as a chain $\{1\} \subset H_2 \subset H_{23}$, is in fact a diamond ($2 \nmid 23$).

1 Introduction

For a prime q let

$$Q_q(n) = n^q - (n-1)^q = \sum_{j=0}^{q-1} n^j (n-1)^{q-1-j}, \tag{1}$$

an irreducible polynomial of degree $q-1$ with leading coefficient q and the symmetry $Q_q(1-n) = Q_q(n)$. For $q = 3$, $Q_3(n) = 3n^2 - 3n + 1$ and its prime values are the cuban primes. We call $n, n+1, \ldots, n+k-1$ a *prime k-tuple* (doublet, triplet, quadruplet, quintuplet for $k = 2, 3, 4, 5$) if $Q_q(n), \ldots, Q_q(n+k-1)$ are all prime.

The Bateman–Horn conjecture [1] predicts the number of such k-tuples with $2 \le n \le Y$ to be

$$N_k(Y) \sim C_k(q) \int_2^Y \frac{dt}{\prod_{i=0}^{k-1} \ln Q_q(t+i)}, \qquad C_k(q) = \prod_p \frac{1 - \omega_k(p)/p}{(1-1/p)^k}, \tag{2}$$

where $\omega_k(p)$ is the number of residues n mod p with $p \mid Q_q(n)Q_q(n+1)\cdots Q_q(n+k-1)$. This paper computes $C_k(q)$, tests (2) on large data sets, and reports constellations and single primes of Q_{47} at sizes up to 200,000 digits.

Related work. Since $n - (n-1) = 1$, $Q_q(n) = \Phi_q(n, n-1)$ is a value of the homogeneous cyclotomic polynomial. The congruence $p \equiv 1 \pmod{q}$ for its prime factors is the classical theory of primitive divisors of $a^q - b^q$ [9, 10]. Prime values of the family have been tabulated for small q: the cuban primes ($q = 3$) [11], the primes $n^5 - (n-1)^5$ [12], and the least b with $\Phi_m(b, b-1)$ prime [13]. We know of no census of consecutive prime values ($k \ge 2$) for $q \ge 5$ apart from the author's earlier preprints on $q = 47$. The Bateman–Horn conjecture [1] is surveyed in [6]. Numerical tests go back to Shanks for $n^2 + 1$ [7]. The singular products converge slowly and only conditionally; Shanks and Lal accelerated them with Dedekind zeta values [8]. We use plain truncated products (Section 3), which is adequate at the precision needed here.

This paper extends the author's earlier preprints on Q_{47}: the constant $C_1(47) \approx 8.68$ [3], the census of primes and constellations for $n \le 2 \times 10^9$ [4], the quadruplet census to 2×10^{11} [5], and the Bateman–Horn verification to 8×10^{11} with the 1000-digit quadruplets [2]. The quadruplet counts given here (749 below 2×10^{11} and 2,393 below 8×10^{11}, each one confirmed by BPSW) supersede the earlier figures of 742 and 2,359, which did not yet include quadruplets found in later sweeps. New here are: the constants $C_k(q)$ for ten exponents and the explanation of their spread; the complete recount of seven exponents, which corrects Version 1; the control search; 21 quintuplets; the 200,000-digit probable prime; the detrended re-examination of the Riemann-zero correlation.

2 Arithmetic of Q_q

Proposition 1. *Let q be an odd prime and p a prime.*

1. *If $p \mid Q_q(n)$ then $p \equiv 1 \pmod{2q}$. In particular $Q_q(n)$ is odd, and its smallest possible prime factor is the least prime $\equiv 1 \pmod{2q}$ (for $q = 47$: 283).*

2. *For $p \equiv 1 \pmod q$, $p \mid Q_q(n)$ holds exactly on the $q-1$ classes $n \equiv (1-u)^{-1} \pmod p$, $u^q = 1 \neq u$; these come in pairs r, $1-r$.*

3. *$Q_q(n) \equiv 1 \pmod{\kappa(q)}$ for all n, where $\kappa(q) = \prod_{(\ell-1)\mid(q-1)} \ell$ is the denominator of the Bernoulli number B_{q-1}; for $q = 47$, $\kappa = 2 \cdot 3 \cdot 47 = 282$.*

Proof. (1)–(2): if $p \mid Q_q(n)$ then $p \nmid n$ (otherwise $Q_q(n) \equiv 1$) and $u = (n-1)/n$ satisfies $u^q \equiv 1$, $u \not\equiv 1$, so u has order q in $\mathbb{F}_p^\times$, $q \mid p-1$, and p is odd since $Q_q(n)$ is odd. Conversely $n = (1-u)^{-1}$ recovers n, and $u \mapsto u^{-1}$ maps n to $1-n$. (3) follows from Fermat's little theorem; see [14]. □

Hence $\omega_1(p) = q-1$ if $p \equiv 1 \pmod q$ and $\omega_1(p) = 0$ otherwise, and $\omega_k(p)$ is the size of the union of the root set R_p and its translates $R_p - 1, \dots, R_p - (k-1)$. Over $\mathbb{Q}(\zeta_q)$ one has $Q_q(n) = N_{\mathbb{Q}(\zeta_q)/\mathbb{Q}}\big(n - \zeta_q(n-1)\big)$, which is the source of these congruences.

Subgroup lattice. Since $(\mathbb{Z}/q\mathbb{Z})^*$ is cyclic of order $q-1$, its subgroup lattice is the divisor lattice of $q-1$. For $q = 47$ it has four elements $\{1\} \subset H_2, H_{23} \subset (\mathbb{Z}/47\mathbb{Z})^*$ with H_2 and H_{23} incomparable. Version 1 used $I(q) = d(q-1) - 2$, the number of proper non-trivial subgroups, as a predictor of clustering; Section 3 shows that it does not predict $C_4(q)$.

3 The Bateman–Horn Constants

The product in (2) converges only conditionally: primes $p \not\equiv 1 \pmod q$ contribute $(1-1/p)^{-k} > 1$, and primes $p \equiv 1 \pmod q$, of relative density $1/(q-1)$, contribute $1 - \omega_k(p)/p$ with $\omega_k(p) \approx k(q-1)$. We evaluated the partial products over all $p \le M$ with $M = 2\times10^6$, 2×10^7, 2×10^8; between $M = 2\times10^6$ and 2×10^8 the values change by less than 0.5% for every q and k. Table 1 lists the values at $M = 2\times10^8$. Our $C_4(47) = 6515$ agrees with the independent value 6514.2 of [2].

Table 1: Bateman–Horn constants (partial products to $p \le 2\times10^8$) and predicted quadruplet counts $N_4(Y)$. p_1 is the least prime $\equiv 1 \pmod q$, and $f_4(p_1) = (1-\omega_4(p_1)/p_1)/(1-1/p_1)^4$ its local factor in C_4. SG: $2q+1$ prime.

q	safe	SG	C_1	C_2	C_3	C_4	p_1	$f_4(p_1)$	$N_4(10^8)$	$N_4(2\cdot10^9)$
23	✓	✓	4.263	17.41	63.05	324	47	0.093	1.53	16.0
29		✓	4.767	23.57	122.3	754	59	0.127	1.36	14.2
37			7.358	55.84	443.6	3749	149	0.407	2.48	26.0
41		✓	5.742	33.85	228.1	1263	83	0.089	0.55	5.76
43			7.710	61.04	464.6	3753	173	0.349	1.35	14.1
47	✓		**8.683**	**77.94**	**721.9**	**6515**	283	0.563	1.63	17.0
53		✓	6.058	37.59	274.4	2003	107	0.107	0.31	3.20
59	✓		9.893	97.99	960.5	9373	709	0.708	0.93	9.70
61			9.170	82.80	770.2	7032	367	0.488	0.61	6.36
71			9.603	92.13	888.9	8532	569	0.614	0.40	4.17

For $q = 47$ we also obtain $C_5 = 58{,}552$ and $C_6 = 535{,}524$.

What determines $C_4(q)$. The constants vary by a factor of 30 across exponents of similar size. The variation is dominated by the least prime $p_1 \equiv 1 \pmod q$. If $p_1 = 2q+1$, the $4(q-1)$ excluded classes nearly exhaust $\mathbb{Z}/p_1\mathbb{Z}$ (e.g. $\omega_4(83) = 76$ for $q = 41$), and the local factor is about 0.1. If $2q+1$ is composite, then $p_1 \ge 4q+1$ and the local factor is 0.35–0.7. Being a safe prime is unrelated: $q = 23$ is both safe and Sophie Germain and has the smallest C_4; $q = 47$ is safe but not Sophie Germain ($95 = 5\cdot19$). The interference potential $I(q)$ of v1 is likewise uncorrelated with C_4: $I = 2$ for $q = 23$ and 47, $I = 6$ for both $q = 41$ ($C_4 = 1263$) and $q = 71$ ($C_4 = 8532$).

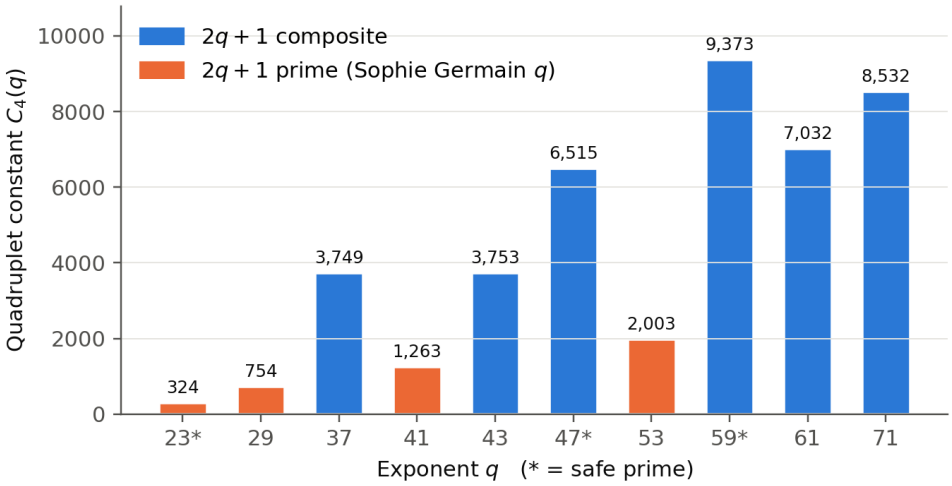


Figure 1: Quadruplet constant $C_4(q)$. The four Sophie Germain exponents ($2q+1$ prime) have the smallest constants, because the prime $p = 2q+1$ excludes almost all residue classes. The safe primes 23, 47, 59 (marked $*$) are spread across the range.

4 Data and Methods

Exhaustive range. For seven exponents $q \in \{37, 41, 43, 47, 53, 61, 71\}$, $Q_q(n)$ was tested for $n \le 2 \times 10^9$ with Miller–Rabin tests [15] (v1).

Q_{47} quadruplet census. Segmented sweeps over $n < 8 \times 10^{11}$ used a sieve on the excluded classes of Proposition 1 followed by probable-prime tests of $Q_{47}(n), \ldots, Q_{47}(n+3)$. All quintuplets were re-checked with BPSW tests [16] in GMP.

Independent recount. To check the census below 2×10^9 independently, we wrote a second program (`quad_census.py`): a segmented sieve with all primes $p \equiv 1 \pmod{q}$, $p \le 3 \times 10^7$, applied to the four shifts, a strong base-2 test of each survivor, and BPSW confirmation of every quadruplet found. This program was run for $q = 37, 41, 43, 47$. For $q = 53, 61, 71$ the recount comes from separate full sweeps of $1 \le n \le 2 \times 10^9$: a sieve with $p \le 5 \times 10^6$ in blocks of 10^7, followed by Miller–Rabin tests. These sweeps skip the last three starts of each block, so those 600 starts per exponent were tested separately (`patch_titan_gaps.py`); none is a quadruplet. Every reported quadruplet was confirmed by BPSW.

Control search. For eight exponents, `control_search.py` counts primes and k-tuples for $n \le 10^7$ after sieving with $p \le 10^5$.

Large windows. Windows of consecutive n starting at 10^{20} and at

$$n_0 = 4797791281515011781318$$

(chosen so that Q_{47} has 1000 digits) were sieved on the excluded classes and searched with probable-prime tests. Quadruplet members at 1000 digits were certified with Primo 4.3.3 [19].

All scripts, logs and certificates are in the repository listed under Data Availability.

5 Single Primes and Tuples of Q_{47} up to 2×10^9

Up to $n = 2 \times 10^9$ there are 18,473,624 primes $Q_{47}(n)$. Bateman–Horn with $C_1 = 8.683$ predicts 18,464,550 ($O/E = 1.0005$). There are 176,896 doublets and 1,779 triplets (overlapping runs counted), against predictions of 176,780 ($O/E = 1.0007$) and 1,752 ($O/E = 1.015$, $z = +0.65$). The prime list consists of the author's search files [4]. One window, $1{,}399{,}874{,}855 \le n \le 1{,}399{,}880{,}000$, was missing from them and was re-tested; it adds 53 primes (`patch_prime_gaps.py`). No other run of more than 2000 consecutive n without a prime occurs.

The author also searched two windows near 10^{44}, where $Q_{47}(n)$ has 2026 digits. In $10^{44} < n \le 10^{44} + 10^6$ there are 1,870 primes against a prediction of 1,861.6 ($z = +0.19$). In a window of 10^7 consecutive n starting at $n \approx 1.004 \times 10^{44}$ there are 18,715 primes against 18,614 ($z = +0.74$). These windows contain 5 and 24 doublets against 3.6 and 35.8 predicted, and no triplets. Both windows were rescanned independently (`rescan_window.py`). Every n was sieved by all $p \equiv 1 \pmod{47}$ with $p \le 10^8$; this left 264,775 and 2,647,699 candidates. These were tested as Fermat base-3 probable primes with PFGW, and each hit was confirmed by BPSW. The rescan reproduces both lists exactly: no prime is missing, and none fails to be confirmed. The 15 quadruplets of v1 (Table 2) compare with a prediction of 17.0.

Table 2: The 15 quadruplets of Q_{47} with $n < 2 \times 10^9$.

k	n	k	n	k	n
1	23,159,557	6	423,600,750	11	1,370,439,187
2	117,309,848	7	523,331,634	12	1,643,105,964
3	136,584,738	8	640,399,031	13	1,691,581,855
4	218,787,064	9	987,980,498	14	1,975,860,550
5	411,784,485	10	1,163,461,515	15	1,996,430,175

6 The Q_{47} Census to 8×10^{11}

Table 3 and Figure 2 compare the quadruplet census with (2). Over the whole range we observe 2,393 quadruplets against a prediction of 2,404.3 ($O/E = 0.995$, $z = -0.23$). No sub-range deviates by more than 1.6 standard deviations.

Table 3: Q_{47} quadruplets by range of the starting point n.

Range of n	Observed	Bateman–Horn	O/E	z
$[2, 2 \times 10^9)$	15	17.0	0.88	-0.49
$[2 \times 10^9, 10^{10})$	45	45.5	0.99	-0.07
$[10^{10}, 3 \times 10^{10})$	77	91.5	0.84	-1.52
$[3 \times 10^{10}, 10^{11})$	256	264.2	0.97	-0.51
$[10^{11}, 2 \times 10^{11})$	356	328.7	1.08	$+1.51$
$[2 \times 10^{11}, 4 \times 10^{11})$	613	591.2	1.04	$+0.90$
$[4 \times 10^{11}, 6 \times 10^{11})$	522	546.5	0.96	-1.05
$[6 \times 10^{11}, 8 \times 10^{11})$	509	519.8	0.98	-0.47
Total	2393	2404.3	0.995	-0.23

Quintuplets and sextuplets. Among the quadruplets, 21 pairs have consecutive starting points, giving 21 prime quintuplets $Q_{47}(n), \ldots, Q_{47}(n+4)$ with $n < 8 \times 10^{11}$ (Table 4); the prediction with $C_5 = 58{,}552$ is 17.9 ($z = +0.73$). No sextuplet occurs; the prediction is 0.14. All 105 quintuplet members pass BPSW.

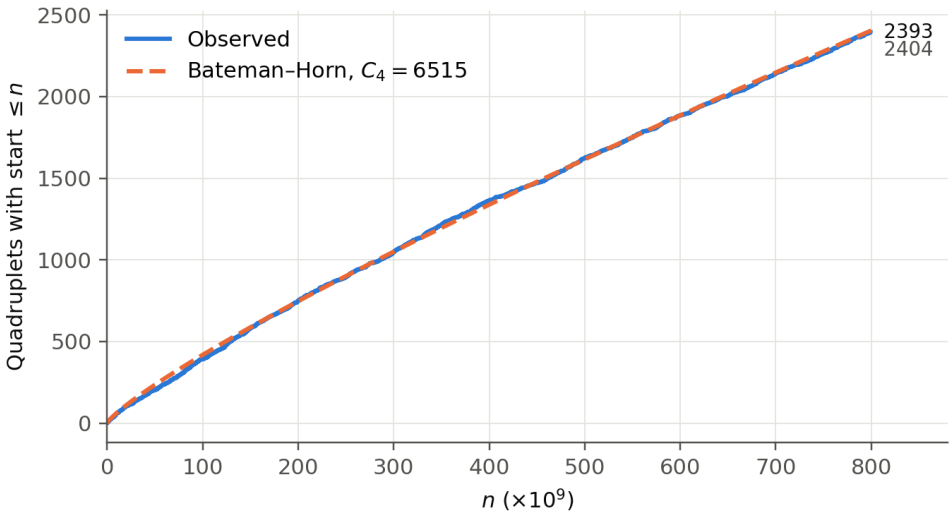


Figure 2: Cumulative number of Q_{47} quadruplets (solid) and the Bateman–Horn prediction with $C_4 = 6515$ (dashed).

Table 4: The 21 prime quintuplets of Q_{47} with $n < 8 \times 10^{11}$ (first element n).

35,676,017,721	201,466,943,077	550,354,512,960
64,482,563,907	204,844,429,912	594,968,850,765
73,292,417,435	254,029,336,327	619,475,391,167
116,255,850,744	305,028,385,174	651,420,729,296
147,743,683,226	342,487,197,658	654,959,149,370
159,430,471,996	350,118,929,558	679,687,730,459
182,501,065,420	362,462,223,971	758,524,005,103

7 Seven Exponents and Control Exponents

Version 1 compared seven exponents using the quadruplet counts in Table 5, and attributed the large count for Q_{47} to the safe-prime property. Table 5 adds the Bateman–Horn predictions for $n \leq 2 \times 10^9$ and the independent recount.

For $q = 47$ the recount finds exactly the fifteen quadruplets of Version 1, at the same coordinates. For the other six exponents, Version 1 reported $2, 0, 1, 0, 0, 0$. The recount finds $23, 3, 10, 2, 4, 6$ for $q = 37, 41, 43, 53, 61, 71$, so the Version 1 counts for these exponents were incomplete. Every recount agrees with Bateman–Horn to within $|z| < 1.2$ ($z = -0.59, -1.15, -1.09, -0.67, -0.94, +0.90$). With complete counts, Q_{37} has more quadruplets below 2×10^9 than Q_{47}, which is what its constant predicts once the different growth rates of the integrand are taken into account. The apparent singularity of Q_{47} in Version 1 came from comparing a complete census against incomplete ones; there is no excess to explain.

Control exponents. Table 6 gives, for $n \leq 10^7$, the observed and predicted numbers of primes, doublets and triplets for eight exponents, including the safe primes 23 and 59 and the Sophie Germain primes 23, 29, 41, 53. For primes and doublets every exponent agrees with Bateman–Horn to within $|z| < 1.6$. Triplets run slightly above prediction in seven of the eight exponents (139 observed against 112.6 predicted in total, $z \approx +2.5$). This mild excess is shared by safe and non-safe exponents alike, and it is largest for $q = 29$, which is not a safe prime. The likely source is the slow, conditional convergence of the truncated constants together with the lower-order terms neglected in the integrand at this small range. The two safe primes 23 and 59

Table 5: Quadruplets with $n \le 2 \times 10^9$: count reported in v1, independent recount, and Bateman–Horn prediction.

q	$I(q)$	v1 count	Recount	BH prediction	C_4
37	7	2	23	26.0	3749
41	6	0	3	5.76	1263
43	6	1	10	14.1	3753
47	2	15	15	17.0	6515
53	4	0	2†	3.20	2003
61	10	0	4†	6.36	7032
71	6	0	6†	4.17	8532

† Full sweep with block-boundary patch (Section 4).

show no surplus at any k. The safe-prime hypothesis of Version 1 is therefore not supported.

Table 6: Control search, $2 \le n \le 10^7$: observed / Bateman–Horn predicted.

q	primes	doublets	triplets	quadruplets
23*	127511 / 127611	1551 / 1570.0	14 / 17.3	0 / 0.3
29	112013 / 112302	1332 / 1316.5	26 / 16.3	2 / 0.2
41	94714 / 94858	955 / 929.8	14 / 10.5	0 / 0.1
43	121537 / 121328	1527 / 1521.4	23 / 18.5	0 / 0.2
47*	125373 / 124809	1680 / 1620.8	26 / 21.9	0 / 0.3
53	77296 / 77067	638 / 612.2	7 / 5.8	0 / 0.1
59*	113068 / 112889	1294 / 1284.1	18 / 14.6	1 / 0.2
71	90788 / 90844	872 / 829.8	11 / 7.7	0 / 0.1

* safe prime.

8 Constellations at 922 and 1000 Digits

Window at $n = 10^{20}$. In the window $[10^{20}, 10^{20} + 3.04 \times 10^{10}]$, where $Q_{47}(n)$ has 922 digits, we found 14 quadruplets. Bateman–Horn predicts 9.8 for the window up to the last quadruplet found; that figure is a lower bound, since the window searched extends beyond it. All 56 members pass BPSW.

Window at 1000 digits. Starting from $n_0 = 4797791281515011781318$, a sweep of at least 3.70×10^{10} consecutive n found the 11 quadruplets of Table 7, against a prediction of at least 8.6. All 44 members pass BPSW. For the first eight, all four members have been proved prime with Primo 4.3.3 (32 ECPP certificates, each taking about one minute of wall-clock time). These are, to our knowledge, the first proven prime quadruplets of a polynomial of degree 46 at 1000 digits. None extends to a quintuplet.

9 A 200,000-Digit Q_{47} Probable Prime

Proposition 2. *Let $n^* = 586135293385 \cdot 10^{4336} + 1647$, an integer of 4,348 digits. Then*

$$P^* = Q_{47}(n^*) = (586135293385 \cdot 10^{4336} + 1647)^{47} - (586135293385 \cdot 10^{4336} + 1646)^{47}$$

has exactly 200,000 decimal digits (664,383 bits) and is a probable prime: it passes the tests in Table 8.

Table 7: The 11 quadruplets of Q_{47} at 1000 digits, $n = n_0 + t$ with $n_0 = 4797791281515011781318$.

#	t	status	#	t	status
1	720,056,643	ECPP	7	17,840,013,419	ECPP
2	7,642,492,197	ECPP	8	28,374,845,686	ECPP
3	9,120,900,759	ECPP	9	34,066,697,501	BPSW
4	12,970,778,218	ECPP	10	36,297,085,036	BPSW
5	15,262,442,996	ECPP	11	37,001,546,293	BPSW
6	16,351,036,649	ECPP			

The decimal expansion of P^* begins and ends with

`1000000000069212334213...` `...71630685699703770614801190644`7,

and the SHA-256 hash of its full decimal string is

`bd23db83e0c523fd4ddbf4a0a885bd09`
`86db93eacf6fd2051e4a887d1f0a8c70`.

Table 8: Verification record for P^*. PFGW 4.1.7 (64-bit, build 20251229) with GWNUM 31.4; GMP 6.3.0 through gmpy2 2.3.0; one run each on a Windows 11 workstation (AMD Ryzen 7 5800H). The GMP rows use independent multiplication code. P^* and $P^* - 1$ both have 664,383 bits; the counter `664382/664382` printed by PFGW is the number of modular squarings in $a^{P^*-1} \bmod P^*$, one less than the bit length of the exponent.

Test	Software	Result	Time
Fermat base 3	PFGW	3-PRP	277.5 + 89.6 s
Fermat base 5	PFGW `-b5`	5-PRP	274.3 + 88.7 s
Fermat base 7	PFGW `-b7`	7-PRP	280.3 s
Fermat base 11	PFGW `-b11`	11-PRP	320.7 s
Strong (Miller–Rabin) base 11	GMP	strong 11-PRP	2642 s
BPSW (strong base 2 + strong Lucas)	GMP	BPSW-PRP	21,839 s
Congruence	Python	$P^* \equiv 1 \pmod{282}$	—

The PFGW tests can be repeated on a recent x86-64 desktop CPU in about five minutes per base, with the input line given under Data Availability. The congruence is the identity of Proposition 1(3); it checks the evaluation of P^*, not its primality. To our knowledge P^* is the largest known probable prime of the form $n^{47} - (n-1)^{47}$.

9.1 Search

Since $Q_{47}(n) = 47n^{46}(1 + O(1/n))$, 200,000 digits require $\log_{10} n \approx (200000 - \log_{10} 47)/46 \approx 4347.79$. We wrote $n = a_0 \cdot 10^{4336} + t$ with $a_0 = 586135293385$, the least 12-digit prefix for which Q_{47} has 200,000 digits, and $0 \le t < 1.5 \times 10^5$. Sieving with all primes $p \equiv 1 \pmod{94}$ up to 2×10^{10} (46 classes each, Proposition 1) left 30,809 candidates (20.5%). These were split into eight equal slices and tested with PFGW in parallel; P^* ($t = 1647$) was the 341st candidate of the first slice.

9.2 Comparison with Bateman–Horn

For candidates free of prime factors $\le B$, the probability of primality is about

$$\frac{e^{\gamma} \ln B}{\ln Q_{47}(n)} = \frac{1.781 \cdot \ln(2 \times 10^{10})}{200000 \ln 10} \approx \frac{1}{10{,}900};$$

the constant C_1 cancels after conditioning on the sieve. The window should contain about 2.8 probable primes. When P^* was found, between 2.7×10^3 and 3.2×10^3 candidates had been tested, for an expected 0.25–0.30 probable primes and a Poisson probability of 22–26% of at least one. The discovery is consistent with Bateman–Horn.

9.3 Primality status

P^* is a probable prime, not a proven prime. An $N-1$ proof [17] needs a factored part of P^*-1 of about one third of its size. With $m = n^* - 1$, the Cauchy–Mirimanoff factorization for $47 \equiv 5 \pmod 6$ gives

$$P^* - 1 = (m+1)^{47} - m^{47} - 1 = 47\, m\,(m+1)\,(m^2+m+1)\,E_{47}(m), \qquad \deg E_{47} = 42,$$

so even complete factorizations of m, $m+1$ and m^2+m+1 would supply about 17,400 of the 200,000 digits (8.7%). ECPP proofs have so far been completed only below 10^5 digits. P^* should therefore be reported as a probable prime, for example to the PRP Top Records list [20].

10 Exploratory Note on Quadruplet Positions and Riemann Zeros

Version 1 reported that the starting points $n_1 < \cdots < n_{15}$ of Table 2 satisfy $n_k^{1/2.74} \approx c\,\gamma_k$ with $r = 0.994$, where γ_k is the k-th ordinate of a zero of $\zeta(s)$, and derived an "effective modulus" $q_{\text{eff}} = e^{2.74}$ from the fitted exponent. We re-examined this in two ways.

Null model. Two increasing sequences correlate strongly whatever their origin. For 2×10^4 sets of 15 positions drawn uniformly from $[10^6, 2 \times 10^9]$ and sorted, the best correlation over power transforms $n^{1/a}$, $a \in [1,5]$, has median 0.985 and 95th percentile 0.994; $r \geq 0.994$ occurs in about 5–6% of sets.

Detrended test. We removed the smooth trend on both sides: $a_k = N_4(n_k) - (k - \frac{1}{2})$ with N_4 from (2), and $b_k = \bar{N}(\gamma_k) - (k - \frac{1}{2})$ with $\bar{N}(T) = \frac{T}{2\pi} \ln \frac{T}{2\pi e} + \frac{7}{8}$. The correlation of the residuals is $r = -0.0003$ (Spearman 0.09), with permutation $p = 0.999$ (Figure 3).

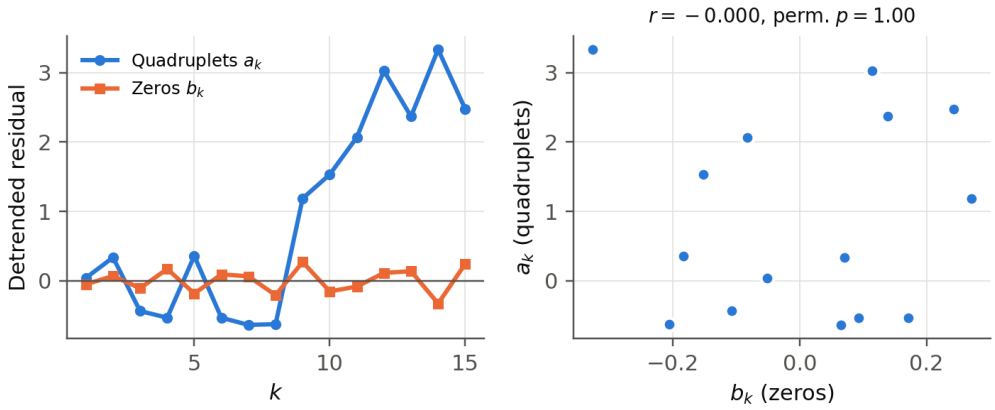


Figure 3: Detrended residuals of quadruplet positions (a_k) and Riemann zero ordinates (b_k). Left: residuals by index. Right: no relation between them.

The quadruplet positions follow the Bateman–Horn trend, and the zero ordinates follow the Riemann–von Mangoldt trend; beyond these trends there is no relation. The fitted exponent

2.74, and with it q_{eff} and the "$q/3$ rule" of v1, describe how two smooth curves can be matched over a short range and carry no arithmetic information. We withdraw these claims.

11 Discussion

At every scale tested—single primes to 2×10^9, quadruplets to 8×10^{11}, quintuplets, and windows at 922 and 1000 digits—the counts for Q_{47} agree with Bateman–Horn within ordinary Poisson fluctuation. The abundance of Q_{47} quadruplets relative to other exponents is a property of the singular series: $C_4(47)$ is large because the smallest prime that can divide $Q_{47}(n)$ is 283, while for $q = 41$ and $q = 53$ the primes 83 and 107 remove almost every residue class. A useful rule of thumb follows: exponents q for which $2q + 1$ is prime are poor sources of constellations of Q_q.

Open problems.

1. A primality certificate for P^*, or a method using the cyclotomic structure of Q_{47} to supply the factored part a proof would need.
2. Completion of the 200,000-digit window (2.8 probable primes expected).
3. Certificates for quadruplets 9–11 of Table 7, and a 1000-digit quintuplet (roughly $(C_5/C_4)/\ln Q \approx 4\times10^{-3}$ quintuplets per quadruplet at that size).
4. A closed-form estimate of $C_k(q)$ in terms of the primes $p \equiv 1 \pmod q$ below a few multiples of q.

Data Availability

This version is archived at Zenodo under the DOI 10.5281/zenodo.23517573. Data, scripts, logs and certificates are available at

https://github.com/Ruqing1963/Polynomial-Prime-Quadruplets

including the constants (`analysis/bh_constants.py`), the census coordinates, the recount and control programs, the Primo certificates, and the 200,000-digit search (`prp200k/`). The PFGW input line

```
(586135293385*10^4336+1647)^47-(586135293385*10^4336+1647-1)^47
```

reproduces P^*.

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